DAVID ADEOGUN

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PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience in model development, training, debugging, optimization, and evaluation across applied ML research and production systems. Published researcher with active work in predictive maintenance (LSTM/AdaBoost), deep learning for remote sensing (Vision Transformers), and AI-driven decision-support frameworks. Strong background in designing experiments, benchmarking model performance, reasoning about ML system design tradeoffs, and communicating technical findings with clarity. Proven ability to bridge research and engineering through end-to-end ML pipelines, from data generation to model deployment and monitoring.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, LSTMs, CNNs, Vision Transformers, AdaBoost, YOLO, Neural Networks, ONNX Runtime, TensorFlow Serving

Experimentation & Evaluation: MLflow, Experiment Tracking, Model Registry, Hyperparameter Tuning, A/B Evaluation, Benchmark Design, RMSE/Precision/Recall Metrics

ML Infrastructure & Ops: AWS (SageMaker, ECS, S3), Kubernetes, CI/CD for ML Pipelines, Automated Rollbacks, Production Monitoring, Docker

Programming & Data: Python, C++, SQL, MATLAB, NumPy, Pandas, Apache Kafka, MQTT, InfluxDB, Azure Stream Analytics

Perception & Robotics: OpenCV, PCL, ROS/ROS2, LiDAR Processing, 3D Point Cloud Segmentation, Sensor Fusion

INDUSTRY EXPERIENCE

Machine Learning Engineer | *Independent Consultant* | Atlanta, GA — Oct 2025 – Present

- Designed and authored comprehensive evaluation suites for machine learning engineering tasks, converting complex research workflows into structured benchmarks using PyTorch and scikit-learn.
- Assessed AI-generated solutions across the ML lifecycle, specifically debugging model training and optimization pipelines using TensorFlow Serving and ONNX Runtime.
- Translated practical ML research into standardized evaluation metrics, utilizing MLflow for experiment tracking and versioning of benchmark results.
- Analyzed ML system design choices and tradeoffs, providing technical reasoning for model selection based on accuracy and resource efficiency across AWS (SageMaker, S3) and Kubernetes environments.
- Engineered evaluation frameworks for LLM reasoning, utilizing schema-driven pipelines and Azure OpenAI (GPT-4o) to benchmark groundedness and decision-logic.
- Developed synthetic data generation pipelines producing 150,000+ labeled cycles with NumPy and Pandas to create controlled benchmarking environments for model robustness.
- Benchmarked temporal vs. non-temporal architectures, using LSTMs and AdaBoost to provide detailed error analysis on sequential data processing capabilities.
- Implemented telemetry-based quality monitoring to track model performance in real-time, leveraging InfluxDB and Azure Stream Analytics to refine evaluation criteria.
- Facilitated rigorous model debugging by integrating Weights & Biases or MLflow to document and evaluate performance across multiple training iterations.
- Communicated complex technical findings with high attention to detail, producing architecture artifacts that defined the end-to-end evaluation topology for frontier models.

MLOps Engineer | *Natterbase* | Atlanta, GA — Oct 2024 – Sep 2025

- Architected end-to-end MLOps infrastructure for enterprise AI applications, building scalable pipelines for data ingestion, model training, CI/CD-driven deployment, and production monitoring on AWS with Kubernetes.
- Implemented model registry, experiment tracking, and automated rollback mechanisms to ensure reproducibility and reliability of ML models in production.
- Integrated a centralized model registry to enforce version control and ensure strict reproducibility of evaluation results for enterprise-grade AI applications.
- Engineered automated rollback mechanisms triggered by performance degradation, ensuring the reliability of ML systems through continuous telemetry and monitoring.
- Built robust data ingestion pipelines utilizing Apache Kafka and MQTT to feed real-time telemetry into evaluation benchmarks for predictive maintenance models.

Cloud Platform Engineer | *American Honda Finance (Contract)* | Alpharetta, GA Feb 2023 – Sep 2024

- Designed and built secure, scalable Azure foundations while automating infrastructure and deployments using modern DevOps practices, managing AKS-based containerized workloads.

Software Engineer | *Williams Sonoma Inc* | Atlanta, GA Jun 2020 – Jan 2023

- Engineered and scaled backend services for catalog, cart, and checkout workflows, supporting high-traffic production systems and ensuring system reliability during peak loads

MACHINE LEARNING RESEARCH EXPERIENCE

Kennesaw State University – *Digital Automation Research Lab*

Advisor: Professor David Guerra-Zubiaga

- Trained and compared LSTM time-series model against AdaBoost baseline classifier for predictive maintenance, achieving earlier fault detection (5,000+ cycle lead time), <5% false alarm rate, and RUL prediction RMSE <2,000 cycles.
- Designed digital twin of ABB IRB 1200 robotic cell in Siemens Tecnomatix Process Simulate, validating IIoT-to-ML data pipeline architecture for Industry 4.0 predictive maintenance evaluation.
- Engineered synthetic IIoT telemetry generation pipeline producing 150,000+ labeled cycles with realistic noise injection and three degradation patterns (linear, exponential, logarithmic) for model training and benchmarking.
- Demonstrated that temporal modeling (LSTM on 500-cycle sequential windows) captures gradual degradation trends missed by snapshot-based approaches, informing ML system design tradeoffs for time-series tasks.

Georgia Southwestern State University – *Intelligent Systems Research*

Advisor: Professor Alexander Yemelyanov

- Architected AI-driven decision-support framework combining LLM reasoning, structured decision logic, and cognitive-behavioral problem-solving to produce explainable recommendations for risk-sensitive scenarios.
- Embedded Azure OpenAI (GPT-4o) as the core personalization engine, designing schema-driven user-profile pipelines to generate grounded, user-specific trade-offs and alternative rankings.
- Productionized and evaluated AI systems on Google Cloud with guardrails, telemetry-based quality monitoring, and cross-platform QA, delivering end-to-end architecture documentation.

PUBLICATIONS & PREPRINTS

[1] Adeogun, D., Vu, K., Alam, W., Quansah, E. "Digital Twin–Driven Predictive Maintenance for Pick-and-Place Robotic Cells Using IIoT Telemetry and Machine Learning." ASME IMECE 2026. (To Be Submitted)

[2] Yemelyanov, A., Adeogun, D. "AI-Powered Cognitive-Behavioral Decision Support for Risk Reduction in Insurance Coverage Selection." AHFE 2026. (Submitted)

[3] Adeogun, D., Sandoval, S., Wulfekuhle, V., Vu, K., Kouwenhoven, F. "Improving Efficiency of Deep-Learning based Vision Transformer for Remote Sensing for Satellite Images." ResearchGate, January 2026.

EDUCATION

M.S. Computer Science | Georgia Southwestern State University

- *Thesis: Design and Implementation of a User-Centered AI-Assisted Decision-Support Framework Under Risk and Uncertainty*

M.S. Intelligent Robotic Systems | Kennesaw State University

- *Relevant Coursework: Machine Learning, Machine Vision, Perception/Navigation/Path Planning, Robot Simulation, 3D Point Cloud Manipulation*

B.S. Software Development | Brigham Young University-Idaho